

KRISH JOSHI

AI Systems Engineer | Frontier Model Evaluations • GPU Acceleration • Agentic Benchmarks

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PROFESSIONAL SUMMARY

AI Systems Engineer specializing in **Frontier Model Evaluation, GPU Acceleration, and Agentic Benchmark Architecture** in Python, C/C++, OpenCL, and Linux. Independent contractor for Handshake (Project Dynamo) architecting 10 multi-tier containerized benchmark environments (Harbor format) evaluating reasoning, code generation, and low-level compute capabilities of frontier models. Strong expertise in deterministic test harness design, GPU kernel quantization (W4A16), dataset verification, and automated evaluation pipelines with zero test flakiness.

WORK EXPERIENCE

AI Benchmark Developer & Systems Evaluator (Independent Contractor)

July 2026 – September 2026

Handshake — Project Dynamo

Remote / Global

- **Architected Multi-Tier AI Evaluation Environments:** Built 10 hardened, reproducible Docker environments (Harbor benchmark format) evaluating frontier models across GPU compute kernels, cryptographic protocols, distributed DAGs, and storage subsystems.
- **Deterministic Automated Quality & Grading Gates:** Engineered comprehensive Python evaluation suites (PyTest, CTRF schema reporting, SHA-256 assertions) incorporating exact IEEE-754 tolerances to grade complex multi-step model solutions with a 100% pass rate.
- **Difficulty Calibration & pass@k Benchmarking:** Designed rigorous dual-trial (pass@2 / pass@5) evaluation gating to evaluate frontier model reasoning and code synthesis against non-trivial combinatorial constraints and edge cases.
- **Runtime Sandboxing & Environment Isolation:** Engineered strict execution boundaries (path traversal guards, CPU/memory quotas) preventing data contamination and ensuring reproducible model benchmarking.

KEY TECHNICAL PROJECTS (PROJECT DYNAMO)

Quantized GEMM GPU Compute Engine for Model Inference

OpenCL C, C/C++, Python, PyTest, SIMT Concurrency

- Engineered a high-throughput **W4A16 matrix multiplication kernel** executing tiled dot products with asymmetric 4-bit packed weights and FP32 accumulators on GPU architectures.
- Eliminated workgroup race conditions using local memory barrier fences and designed boundary-safe bit-shift unpacking logic for dynamic, non-power-of-two matrix shapes.
- Built an automated Python-OpenCL evaluation harness verifying numerical accuracy against FP64 reference baselines across ragged multi-dimensional tensors.

Distributed Execution Graph & DAG State Reconciliation Engine

Python, Graph Algorithms, SQLite, PyTest, JSONL

- Architected an algorithmic evaluation benchmark to test model reasoning on 3,500+ line cluster execution logs, resolving node rollbacks and stateful resume graphs.
- Implemented recursive tree pruning, cycle detection, and deterministic tie-breaking algorithms to derive authoritative execution metrics with $\pm 1.0s$ wall-clock accuracy.
- Packaged the engine into a sandboxed evaluation runtime with deterministic PyTest test fixtures verifying edge-case DAG topologies.

3D Rotation Constraint Satisfaction Solver Benchmark

Python, NumPy, Discrete Mathematics, CSP Backtracking

- Designed a combinatorial AI benchmark task modeling spatial orientation constraints over the 24-element chiral octahedral rotation group $SO(3)$.
- Implemented an optimized depth-first search solver with forward checking, non-commutative sliding composition paths, and matrix trace parity verification.
- Built automated verification testing model reasoning across large search spaces requiring intelligent branch pruning.

TECHNICAL SKILLS

- **AI & Evaluation:** AI Benchmark Architecture (Harbor format), LLM Evaluation & Validation, pass@k Metrics, Dataset Verification, Test Sandboxing
- **Languages & Frameworks:** Python (PyTest, NumPy, Pandas, AsyncIO), C/C++, OpenCL C, Bash Scripting, SQL (PostgreSQL, SQLite)
- **High-Performance Computing:** GPU Acceleration (SIMT), Concurrency (SIMD/Multiprocessing), Quantization (W4A16), IEEE-754 Precision Testing
- **Developer & Testing Tools:** Docker Containerization, Linux System Administration, Git & GitHub, PyTest, CTRF Reporting, GCC/Clang

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Computer Science & Engineering

July 2024 – June 2028 (Expected)

Institute of Advanced Research (IAR) — Gandhinagar, India

- **Google Cybersecurity Professional Certificate** — Coursera / Google (Python Automation, Linux, SQL, Network Security)
- **Google Cloud Fundamentals: Core Infrastructure** (Google Cloud) | **Introduction to Networking** (NVIDIA) | **Git & GitHub** (IBM)
- **Industry Job Simulations:** AWS (Solutions Architecture), Accenture Nordics (Software Engineering), Mastercard (Technology)